

Name: _____

Date: _____

Why Do Leaves Change Color?

Grade 3 reading comprehension · Informational text · 11 points · About 15 minutes

Green leaves

In spring and summer, most leaves are green. They are green because they contain a pigment called chlorophyll. A pigment is something that gives color. Chlorophyll helps a leaf turn sunlight into food for the tree.

Changes in fall

In the fall, the days get shorter and the nights get cooler. The tree makes less chlorophyll. Soon the green color fades away.

Hidden colors

Other colors were in the leaf all along! Yellow and orange pigments are called carotenoids. They are the same pigments that make carrots orange. We cannot see them in summer because the green is so strong. When the green fades, the yellow and orange show through.

Red leaves

Some leaves turn red or purple. These colors come from pigments called anthocyanins. A tree makes more of them in the fall, especially when the days are sunny and the nights are cool.

Falling leaves

Before a leaf falls, the tree seals the spot where its stem was attached. This keeps the tree safe through the cold winter. At last, the leaves dry up and fall to the ground. In spring, the tree grows new green leaves, and the cycle begins again.

1. What gives most leaves their green color? (1 point)

2. What does chlorophyll help a leaf do? (1 point)

3. According to the text, what is a pigment? (1 point)

4. Name two things that happen in the fall that change the color of leaves. (2 points)

5. Why can't we see the yellow and orange colors in summer? Use the text to explain. (2 points)

6. Which pigments make some leaves turn red or purple? (1 point)

7. The text says carotenoids are in some leaves. What is one other thing that has the same pigments? (1 point)

8. Write one sentence that tells the main idea of the whole text. (2 points)

Writing task: Think of a tree you know. Write five sentences about how it looks in each season.
